

tinygrad: a single dialect from Tensor programs to Command Buffers

tinygrad, Corp.
research@tinygrad.org

UOps

All nodes in the tinygrad graph are **UOps**. A UOp is a tuple (op, src, arg, tag) where op is from the set below, src is a tuple of input UOps, arg is op-dependent, and tag is for temporary processing. The full program is a DAG of UOps. Each UOp has five derived properties — **dtype**, **shape**, **device**, **addrspace**, and **min_max** — determined by the rules at the end of this document.

Source Ops — leaf nodes

Op	src	arg	Semantics
PARAM	(s)	slot, dtype, device?, addrspace?	Placeholder with shape s . Substituted in FUNCTION.
BUFFER	(s)	slot, dtype, device, addrspace	Concrete buffer slot with shape s . If device is a tuple, it creates the fully sized buffer across multiple devices.
CONST	()	value, dtype	A scalar constant with shape (). Form vector consts with STACK
BINARY	()	data	Raw binary data, has dtype uint8 and shape len(data)

addrspace is GLOBAL, LOCAL, or REG.

Movement Ops — no arithmetic; view, indexing, and reinterpretation only

Op	src	arg	Semantics
PERMUTE	(T,)	axis order π	Reorder axes. $\pi = (1, 0)$ is transpose.
FLIP	(T,)	bools f	Reverse along flagged axes.
RESHAPE	(T, s')	—	Reinterpret in row-major order. $\prod s_k = \prod s'_k$.
EXPAND	(T, s')	—	Broadcast size-1 axes. $s_k \in \{1, s'_k\}$.
PAD	(T, o, s')	—	Place <i>T</i> at offset o_k in an invalid-filled output of shape s'_k .
SHRINK	(T, o, s')	—	Keep s'_k elements starting at offset o_k per axis. Inverse of PAD.
INDEX	(T, i ₀ , i ₁ , ...)	—	Index from left. ()-shaped <i>i</i> removes dim; (<i>k</i> ,)-shaped makes it <i>k</i> .
STACK	(T ₀ , T ₁ , ...)	—	Join along a newly created leading axis. All shapes must match.
BITCAST	(T,)	dtype	Reinterpret storage as target dtype; preserve total bytes.

Reduce Ops — remove axes

Op	src	arg	Semantics
REDUCE	(T, r ₀ , r ₁ , ...)	op, <i>n</i>	Reduce the first <i>n</i> axes of <i>T</i> . Op is ADD, MAX, or MUL.

Call Ops — function abstraction, like the lambda calculus

Op	src	arg	Semantics
FUNCTION	(body, $a_0, a_1, \dots$)	—	Substitute each PARAM k in TUPLE body with a_k . Gradient-able.
CALL	(body, $a_0, a_1, \dots$)	—	Opaque invocation of a compiled kernel or custom function.
TUPLE	($v_0, v_1, \dots$)	—	Pack values; required as FUNCTION body to return a value.
GETTUPLE	($T,$)	idx	Extract element at idx from a TUPLE.

Load Ops — can change device or addrspace

Op	src	arg	Semantics
LOAD	(buf, alt?, gate?)	device, addrspace	Read (pull) from buffer into a new anonymous buffer. Note: this replaces COPY and CONTIGUOUS.

Store Ops — the only op with observable side effects

Op	src	arg	Semantics
STORE	(buf, val, gate?)	—	Write (push) val into buf. buf.shape = val.shape. If gate is present, write only when gate is true. Output is void.

Ordering Ops — execution order

Op	src	arg	Semantics
RANGE	(bound,)	type	Iterator from 0 to bound.
END	(body, range)	—	Close a RANGE loop.
AFTER	(buf, deps...)	—	Passthrough of buf; guarantees deps execute first.
GROUP	($u_0, u_1, \dots$)	—	Void no-op that merges multiple STORES into one node, unordered.
SINK	($s_0, s_1, \dots$)	—	Collect side effects into a single root node.
LINEAR	(uops...)	—	Linearized (toposorted) instruction sequence.

Assign is STORE followed by AFTER: write the value, then return the buffer with an ordering dependency.

Elementwise Ops — all inputs same shape, output same shape, applied per-element

Arity	src	Op	Semantics
Unary	(T,)	RECIP	$1/x$
		TRUNC	$\text{trunc}(x)$: round toward zero.
		CAST	Convert to target dtype (specified in arg).
Binary	(A, B)	ADD, MUL, MAX, MOD, IDIV	$a + b, a \cdot b, \max(a, b), a \bmod b, \lfloor a/b \rfloor$
		CMP<LT, CMP<NE	$[a < b], [a \neq b]$
		XOR, OR, AND, SHR, SHL	$a \oplus b, a \mid b, a \& b, a \gg b, a \ll b$
Ternary	(P, A, B)	WHERE	$A[i]$ if $P[i] \neq 0$, else $B[i]$

Decomposed elementwise ops — defined in terms of the primitives above.

Op	Decomposition	Semantics
NEG	MUL($A, -1$)	$-x$
SUB	ADD($A, \text{NEG}(B)$)	$a - b$
DIV	MUL($A, \text{RECIP}(B)$)	a/b
CMPGT	CMPLT(B, A)	$[a > b]$
CMPGE	CMPNE(CMPLT(A, B), 1)	$[a \geq b]$
CMPLE	CMPNE(CMPLT(B, A), 1)	$[a \leq b]$
CMPEQ	CMPNE(CMPNE(A, B), 1)	$[a = b]$
NOT	CMPNE($A, 1$)	$\neg a$
EXP2	polynomial approx + MUL, ADD	2^x
LOG2	exponent extract + polynomial approx	$\log_2 x$
SIN	argument reduction + polynomial approx	$\sin x$
SQRT	EXP2($0.5 \cdot \text{LOG2}(A)$)	$\sqrt{x}$
POW	EXP2($\text{LOG2}(A) \cdot B$)	a^b
MULACC	ADD(MUL(A, B), C)	$a \cdot b + c$
THREEFRY	5 rounds of add-rotate-xor (ARX)	Threefry 2x32 PRNG

Marker Ops — identity on data

Op	src	arg	Semantics
CONTIGUOUS	(T ,)	—	Force contiguous memory layout.
CONTIGUOUSBACKWARD	(T ,)	—	Force contiguous in backward pass.
DETACH	(T ,)	—	Stops gradient propagation.

Codegen Ops — generated code primitives, these do not appear in the main graph

Op	src	arg	Semantics
BARRIER	(deps...)	—	Synchronize threads within a workgroup.
INS	...	...	A single machine instruction (e.g. AMD ISA).
GETADDR	(buf,)	dev	Lower buf to its address on device dev.
SPECIAL	(bound,)	name	GPU thread/workgroup index (e.g. <code>gidx0</code> , <code>lidx1</code>).
IF	(gate,)	—	Begin conditional execution block.
ENDIF	(if,)	—	End conditional execution block.
WMMA	(A, B, acc)	config	Warp matrix multiply-accumulate (tensor cores).
CUSTOM	(args...)	fnt	Inject custom code string into generated source.
ATOMICADD	(idx, val)	—	Atomic read-modify-write: <code>buf[idx] += val</code> .
CUSTOMFUNCTION	(meta...)	name	Opaque device function (e.g. HW decode). Via <code>CALL</code> .
PROGRAM	(linear, source, binary)	—	Compiled kernel: instructions, source, and machine code.
SOURCE	()	str	Human-readable rendered source code.
BINARY	()	bytes	Compiled machine code.

These ops are not part of the core specification and are subject to change.

Derived Properties

Every UOp has a **dtype**, **shape**, **device**, **addrspace**, and **min_max**, derived from its op, src, and arg:

Op	dtype	shape	device	min_max
BUFFER	from arg	from src[0]	from arg	dtype range
CONST	from arg	()	NULL	$[v, v]$
PARAM	from arg	from src[0]	from arg	from src or dtype range
Movement ops	src[0].dtype	(see op)	src[0].device	src[0]
REDUCE	src[0].dtype	remove first n axes	src[0].device	dtype range
CAST	from arg	src[0].shape	src[0].device	clamped to dtype
BITCAST	from arg	src[0].shape	src[0].device	dtype range
COPY	src[0].dtype	src[0].shape	from arg	src[0]
ALU unary	src[0].dtype	src[0].shape	src[0].device	dtype range
ADD	src[0].dtype	broadcast	src[0].device	$[a + b, A + B]$
MUL	src[0].dtype	broadcast	src[0].device	$[\min, \max]$ of products
MAX	src[0].dtype	broadcast	src[0].device	$[\max(a, b), \max(A, B)]$
Other binary	src[0].dtype	broadcast	src[0].device	dtype range
CMPLT, CMPNE	bool	broadcast	src[0].device	from intervals
WHERE	src[1].dtype	broadcast	src[0].device	$[\min(b, c), \max(B, C)]$
FUNCTION, CALL	src[0].dtype	substitute PARAM shapes	src[1].device	dtype range
RANGE	index	()	NULL	$[0, n-1]$
INDEX	src[0].dtype	remaining dims	src[0].device	src[0]
STORE	void	()	src[0].device	—
AFTER	src[0].dtype	src[0].shape	src[0].device	src[0]

broadcast: right-align shapes, element-wise max; each axis must be equal or 1. $[a, A]$, $[b, B]$, $[c, C]$ denote $\min_max$ of src[0], src[1], src[2]. Default *dtype range*: $[\text{dtype_min}, \text{dtype_max}]$.

axis tracks the multi-device sharding dimension. BUFFER with n -tuple device: $\text{axis} = 0$ (device dim). RESHAPE remaps axis to preserve the shard boundary. PERMUTE follows the permutation. REDUCE on the shard axis $\rightarrow$ NULL (shard axis is among the first n axes). REPLICATED on the shard axis $\rightarrow$ NULL. COPY $\rightarrow$ NULL. ALU ops inherit from sources. Default: NULL.

Kernel Optimizations (OptOps) — schedule-level transforms on kernel ranges

Each kernel's iteration space is a set of RANGE axes. Every range has an **AxisType**:

AxisType	Letter	Split from	Direction	Semantics
GLOBAL	g	—	—	GPU global workgroup dimension.
LOCAL	l	g, L	inner	Workgroup local dimension (shared memory).
WARP	w	(created by TC)	—	Warp-level lanes for tensor cores.
THREAD	t	g	outer	CPU thread parallelism.
LOOP	L	—	—	Generic sequential loop (initial state).
REDUCE	R	—	—	Reduction axis.
GROUP_REDUCE	G	R	inner/outer	Shared-memory group reduction.
UPCAST	u	g, l, L	inner	Register-level vectorization.
UNROLL	r	R, G	inner	Fully unrolled loop.

An optimization is a triple (op, axis, arg):

OptOp	axis	arg	Semantics
SPLIT	any	(factor k , target, top?)	Split axis n by k into $(n/k, k)$ or $(k, n/k)$ if top. New sub-axis gets target AxisType (see table above).
PADTO	any	multiple m	Pad axis to next multiple of m with validity masks.
SWAP	axis_i	axis_j	Swap two axes $i \leftrightarrow j$.
NOLOCALS	—	—	Disable local memory; no workgroup dims emitted.
TC	reduce idx	(tc, opt, mode)	Apply tensor core WMMA: split reduce/output axes into WARP, UPCAST, and UNROLL dims.

Optimizations compose left-to-right. TC must be first. The search space is explored by BEAM search or hand-coded heuristics.

Common Ops as Compositions

All high-level tensor operations decompose into the primitives above.

```
# gemm:  $C[M,N] = A[M,K] @ B[K,N]$ 
def gemm(A, B):
    M,K = A.shape; _,N = B.shape
    return (A.reshape(M,K,1) * B.reshape(1,K,N)).sum(1)

# prefix_sum: cumulative sum via repeat+reshape sliding window trick
def prefix_sum(T):
    n = T.shape[0]
    x = T.pad((n-1, 0))
    x = x.reshape(1,2*n-1).expand(n+1,2*n-1)
    x = x.reshape((n+1)*(2*n-1)).shrink_to(2*n*n)
    x = x.reshape(n,2*n).shrink_to(n,n)
    return x.sum(-1)

# arange: prefix_sum of all 1s gives  $[1,2,\dots,n]$ , subtract 1 for  $[0,1,\dots,n-1]$ 
def arange(n):
    return prefix_sum(Tensor(1).reshape(1).expand(n)) - 1

# gather:  $\text{out}[i] = T[\text{idx}[i]]$ . one-hot mask along gather axis, then reduce
def gather(T, idx):
    K = T.shape[0]
    pos = arange(K).reshape(K, 1)
    mask = (pos == idx.reshape(1, -1)).cast(T.dtype)
    return (T.reshape(K, 1) * mask).sum(0)

# scatter_add:  $T[\text{idx}[i]] += \text{val}[i]$ 
def scatter_add(T, idx, val):
    K, D = T.shape[0], idx.shape[0]
    pos = arange(K).reshape(K, 1)
    mask = (pos == idx.reshape(1, D)).cast(T.dtype)
    return T + (mask * val.reshape(1, D)).sum(1)
```

Multi-Device Collectives — derived from primitives

Let $D = (d_0, \dots, d_{n-1})$ be an n -tuple device. COPY to an n -tuple device reshards with axis = 0. COPY never changes shape.

T has shape $(s,)$ on a single device.

```
# broadcast: replicate  $T$  to all  $n$  devices
def broadcast(T):
```

```

    return T.reshape(1, s).expand(n, s).copy(D).replicated(0)          # (s,) on D, axis=null

# scatter: split T into n chunks, one per device
def scatter(T):
    return T.copy(D)                                                  # (s,) on D, axis=0

# T has shape (n*s,) on D with axis=0, so each device holds (s,) elements.

# gather: collect all shards onto one device
def gather(T):
    return T.copy(D[0])                                              # (n*s,) on D[0], axis=null

# reduce: gather + sum
def reduce(T):
    return gather(T).reshape(n, s).sum(0)                            # (s,) on D[0], axis=null

# allgather: collect all shards, replicate to all devices
def allgather(T):
    return T.reshape(1, n*s).expand(n, n*s).copy(D).replicated(0)   # (n*s,) on D, axis=null

# reduce_scatter: reduce across devices, scatter result
def reduce_scatter(T):
    return T.reshape(n, n, s//n).permute(1, 0, 2).copy(D).sum(1).reshape(s) # (s,) on D, axis=0

# allreduce: reduce_scatter + allgather
def allreduce(T):
    return allgather(reduce_scatter(T))                             # (s,) on D, axis=null

```

The @function Decorator — graph capture via tracing

The @function decorator transforms a Python function on Tensors into a single FUNCTION node.

```

@function
def f(a: Tensor, b: Tensor) -> Tensor:
    return a + b

```

When $f(x, y)$ is called, the decorator:

1. **Extracts inputs:** walks all arguments to find every Tensor, deduplicates by identity.
2. **Runs the function lazily** (no device execution), building a UOp graph from the result.
3. **Parameterizes:** replaces each input UOp with a $\text{PARAM}(k)$ placeholder.
4. **Wraps the body** in a **TUPLE** (even for single returns) and creates $\text{FUNCTION}(\text{TUPLE}(\text{body}), x, y)$.
5. **Returns** the result via $\text{GETTUPLE}(0)$, or one GETTUPLE per element for tuple returns.

The result is a reusable graph fragment: the body contains only PARAM references, not concrete buffers. At schedule time, the **FUNCTION** is resolved by substituting each $\text{PARAM}(k)$ back with its corresponding argument a_k , or lowered into an opaque **CALL** if it is to be compiled as a reusable kernel.

Lowering Pipeline — from Tensor graph to machine code

Stage	Semantics
Callify	Transform the Tensor graph into a single stateless function.
Rangeify	Determine the kernel split of the function. Break everything down to shape ()
Optimize	Insert local buffers. Swap and split ranges, and determine which axes are parallel and which are serial.
Expand	Expand the parallel ranges into shape.
Instruction Selection	Select target instructions, including WMMA and devectorization.
Linearize	Topologically sort the graph and determine execution order.
Register/Memory Plan	Allocate and reuse GLOBAL, LOCAL, and REG storage for values with non-overlapping lifetimes.
Render	Output the machine code.